AP-E7 Discrete Re-relaxation Audit: Finite-site Topology, an Admissibility Obstruction, and Same-grid Sparse Operators

Route F research note

17 July 2026

Abstract

The AP-E6 continuum hedgehog is used only as an initial condition. We then minimize the same declared finite-difference $n+s$ energy independently on five cubic grids with exact-vacuum Dirichlet data. This exposes a structural obstruction: at finite site number the unconstrained configuration space is path-connected, so it has no exact $B \in \pi_3(S^3)$ components. A Freudenthal-tetrahedron regular-value degree equals $B=1$ only on an admissible open subset. Unconstrained solves cross its dislocation boundary and converge to $B=0$. Endpoint-guarded line searches at three explicit admissibility floors have $B=1$ at every accepted iterate, but hit the floor with nonzero projected gradient instead of finding an interior stationary point. Continuous admissibility of the entire geodesic segment between two accepted endpoints is not certified.

For numerical and algebraic controls only, a central 3^3 hedgehog core is fixed and every other variable is re-relaxed. In that artificial sector we assemble an exact sparse constraint-projected $n+s$ Hessian and same-grid diquark, gauge, ghost, and Wilson/Yukawa normal operators. The latter is a positive fermion proxy, not the bosonic second derivative of $-\log\det D_W$. Therefore no genuine unanchored relaxed $B=1$ lattice solution, physical super-Hessian, four-dimensional dynamics, or continuum stability theorem is claimed.

Contents

1 Declared finite-difference problem

Let $n_x \in S^3 \subset \mathbb{R}^4$, $s_x \in \mathbb{R}$, and freeze every boundary site to $n_x = (1, 0, 0, 0)$, $s_x = 1$. The AP-E6 energy is discretized as

$$\begin{aligned} E_a[n, s] = {} & \frac{a}{2} \sum_{\langle xy \rangle} \left(\|n_y - n_x\|^2 + (s_y - s_x)^2 \right) \\ & + \frac{\kappa a^3}{2} \sum_c \sum_{i<j} \left(\|u_i^c\|^2 \|u_j^c\|^2 - (u_i^c \cdot u_j^c)^2 \right) \\ & + a^3 \sum_{x \in \mathrm{int}} \left[\frac{m_s^2}{2} (s_x - 1)^2 + \beta^2 s_x^2 (1 - n_{x0}) \right], \end{aligned} \tag{1}$$

where $u_i^c = (n_{c+\hat{\imath}} - n_c)/a$ and

$$\kappa = 1, \qquad \beta = \frac{1}{2}, \qquad m_s = \frac{5}{2}. \tag{2}$$

This is an ordinary classical three-dimensional regulator of the continuum Skyrme-type model [1]. It is neither a Euclidean four-dimensional measure nor a two-colour lattice-QCD action. Recent two-colour sigma-model studies motivate additional scalar channels but do not remove this distinction [2, 3].

Each grid is initialized by the same AP-E6 $R = 16$ continuum solution with canonical little-endian float64 checksum

`81104f59a5f3f4337739fc2a217cafc8da91581c9f1fb40b4fdba6ce0f948d59`.

The sampled outer faces are replaced by the exact vacuum before each independent lattice solve.

2 Why a finite-site B sector is not automatic

Theorem 2.1 (Finite-site no-sector theorem). *For $M = (N-2)^3 < \infty$ dynamical sites and fixed boundary, the unconstrained configuration space*

$$\mathcal{C}_N = (S^3)^M \times \mathbb{R}^M \tag{3}$$

is path-connected. Consequently no integer-valued continuous function on all of $\mathcal{C}_N$ can distinguish a $B = 1$ component from the vacuum.

Proof. Both S^3 and $\mathbb{R}$ are path-connected. Given two configurations, choose a path between the two values at every site and take their finite Cartesian product. This constructs a path in $\mathcal{C}_N$. A continuous map from a connected space to the discrete space $\mathbb{Z}$ is constant. Thus continuum $\pi_3(S^3)$ cannot become an exact decomposition of the unrestricted finite-site space without an additional admissibility condition, topology-preserving interpolation, or infinite dislocation barrier. □

This is the finite-site form of the dislocation problem familiar from geometric lattice charge [4, 5]. Lattice Skyrme constructions that retain metastable topology introduce an explicit spherical-tetrahedron prescription and stabilizing structure [6].

3 Freudenthal regular-value degree

Split every cube into the six Freudenthal tetrahedra indexed by permutations $p \in S_3$, with vertices

$$x_0 = c, \quad x_1 = c + e_{p_1}, \quad x_2 = c + e_{p_1} + e_{p_2}, \quad x_3 = c + (1, 1, 1). \tag{4}$$

Their domain orientation is $\epsilon(p) = \mathrm{sgn}(p)$. For a generic regular target $q \in S^3$, form

$$V_T = [n_{x0}\ n_{x1}\ n_{x2}\ n_{x3}], \qquad c_T = V_T^{-1} q \quad \text{when } \det V_T \neq 0. \tag{5}$$

There are finitely many tetrahedra. If V_T is singular, its column span is a proper linear subspace of $\mathbb{R}^4$; choose the generic regular target q outside the finite union of those spans. Such a tetrahedron then has no preimage of q and contributes zero. Equivalently, an exceptional target is defined by an arbitrarily small regular perturbation before taking the signed count. Thus the inverse in (5) is never applied to a singular matrix. The numerical implementation uses the declared $|\det V_T| < 10^{-13}$ exceptional-cell tolerance and verifies target agreement and a much larger nonzero determinant margin on every counted preimage. The ray through q intersects the positive cone of the four vertices iff every component of c_T is positive. In that case the local map degree is

$$d_T(q) = \epsilon(p)\, \mathrm{sgn} \det V_T. \tag{6}$$

Thus, in the sign convention of the continuum AP-E6 card,

$$\deg(n; q) = \sum_T d_T(q), \qquad B_{\mathrm{geom}} = -\deg(n; q). \tag{7}$$

Three generic targets are used, and agreement is required.

The normalized affine map on T is defined only if the vertex convex hull does not contain the origin. Its exact margin is

$$\delta_T = \min_{\lambda_i \geq 0,\ \sum_i \lambda_i = 1} \left\| \sum_{i=0}^{3} \lambda_i n_{x_i} \right\|, \qquad \delta_a = \min_T \delta_T. \tag{8}$$

The implementation enumerates all active faces of each four-vertex convex program. When $\delta_a > 0$, normalization is nonsingular and B_{geom} is locally constant. At $\delta_a = 0$, a dislocation can change the integer without contradicting the theorem above. A regression uses the four regular-tetrahedron vectors $(1, 1, 1, 0)$, $(1, -1, -1, 0)$, $(-1, 1, -1, 0)$, and $(-1, -1, 1, 0)$, divided by $\sqrt{3}$. Their equal-weight sum vanishes; the active-set KKT solver returns $\delta_a = 2.21 \times 10^{-16}$ and correctly rejects admissibility.

4 Riemannian relaxation and its audit

At every site choose an orthonormal tangent frame $E_x \in \mathbb{R}^{4\times 3}$. A fixed-base exponential chart is

$$n_x(z_x) = \cos r_x\, n_x^{(0)} + \frac{\sin r_x}{r_x} E_x z_x, \qquad r_x = \|z_x\|, \tag{9}$$

while $s_x = s_x^{(0)} + \sigma_x$. If $u = z/r$, the exact pullback of an embedding gradient $g \in \mathbb{R}^4$ is

$$\begin{aligned} J(z)^T g = u\big[(-\sin r\, n^{(0)} + \cos r\, Eu) \cdot g\big] \\ + \frac{\sin r}{r}(I - uu^T)E^T g, \end{aligned} \tag{10}$$

with limit $E^T g$ at $r = 0$. A centered directional derivative checks Eq. (10) independently; its best relative residual is 7.31×10^{-11}.

Three logically different solves are reported:

(a) unconstrained L-BFGS relaxation of every interior variable;

(b) Riemannian Armijo descent that rejects any candidate endpoint with $B_{\mathrm{geom}} \neq 1$ or $\delta_a < \delta_{\mathrm{floor}}$;

(c) an artificial control fixing the n values, but not s, on the central $3 \times 3 \times 3$ sampled hedgehog block.

Only (a) tests full-grid stationarity. Case (b) diagnoses the admissible boundary. Case (c) tests a deliberately restricted solver and cannot be called a genuine relaxed lattice soliton.

5 Exact restricted Hessian and same-grid blocks

Let $G_x = \partial E_a/\partial n_x$ and $\lambda_x = n_x \cdot G_x$. With $Q = \mathrm{diag}(E_1, 1, E_2, 1, \dots)$, the exact Riemannian second variation is

$$H_{n+s} = Q^T \left[H_{\mathrm{emb}} - \mathrm{diag}(\lambda_1 I_4, 0, \lambda_2 I_4, 0, \dots)\right] Q. \tag{11}$$

For artificial control (c), rows and columns belonging to the 81 fixed core tangent variables are deleted. This restriction is an exact Hessian of Eq. (1) on that constrained manifold. It is not a full-grid physical Hessian. On every grid, an independent centered geodesic second difference with all fixed-core tangent entries set to zero reproduces $v^T H_{n+s} v$; the largest best-step relative residual is 2.44×10^{-9}.

On the identical interior Dirichlet grid, define

$$H_\Delta = I_2 \otimes \left[L_D + m_\Delta^2 + \chi(1 - n_0)\right], \qquad m_\Delta^2 = 0.81, \qquad \chi = -0.30, \tag{12}$$

$$H_A = I_3 \otimes L_D, \qquad H_{\bar{c}c} = L_D, \qquad (\xi = 1). \tag{13}$$

These are declared quadratic blocks. Here H_A has three real spatial components and $H_{\bar{c}c}$ one complex ghost: they are Abelian Feynman-gauge proxies [7]. A physical SU(2) regulator would instead require at least $3 \times 3M = 9M$ real spatial-adjoint gauge variables and $3M$ complex adjoint ghosts, plus interacting cross blocks. Those are absent.

For an explicit same-grid fermion proxy, let

$$\begin{aligned} D_W = \sum_{i=1}^{3} \gamma_i \frac{T_i^+ - T_i^-}{2a} + m\mathbf{1} + \frac{r_W}{2a} \sum_{i=1}^{3} (2 - T_i^+ - T_i^-) \\ + y\left(P_R \otimes U_x + P_L \otimes U_x^\dagger\right), \qquad U_x = n_{x0}\mathbf{1} + i n_{xa}\tau_a, \end{aligned} \tag{14}$$

with $m = 0.60$, $y = 0.80$, $r_W = 1$, and Dirichlet shifts. This is a Wilson/Yukawa operator in the sense of an explicitly declared regulator [8]. We diagonalize

$$K_F = D_W^\dagger D_W \succeq 0. \tag{15}$$

The displayed same-grid direct sum is therefore

$$\mathcal{H}_{\mathrm{declared}} = H_{n+s}^{\text{core restricted}} \oplus H_\Delta \oplus H_A \oplus H_{\bar{c}c} \oplus K_F. \tag{16}$$

Only the first block is a bosonic Hessian of the interacting background energy. H_Δ is the Hessian of a separately declared quadratic sector; H_A is a free gauge proxy; $H_{\bar{c}c}$ is a Grassmann operator; and K_F is a positive normal proxy. Equation (16) is not the BRST super-Hessian nor the second variation of $-\log\det D_W$.

6 Numerical results

The deterministic card records five independently re-relaxed grid/volume pairs:

$$(N, L) = (7, 6), (9, 6), (11, 6), (9, 8), (11, 10). \tag{17}$$

The first three vary $a = L/(N-1)$ at fixed volume; $(7, 6)$, $(9, 8)$, and $(11, 10)$ give the fixed-$a = 1$ volume sequence. No result is obtained by interpolating a previously relaxed lattice field.

6.1 Unconstrained unwind

Table 1 shows that full-grid solves converge to the exact vacuum within numerical tolerance. Here g_a is the sitewise tangent-gradient RMS divided by a^3.

Fixing only the central value $n(0) = -1$ also fails. It converges to $E = 12.46071222$, $g_a = 7.15 \times 10^{-9}$, but all three regular targets give $B_{\mathrm{geom}} = 0$ and $\delta_a = 0$. The single pinned point is surrounded by a degenerate tetrahedral dislocation.

Table 1: Independent full-grid minimizations of Eq. (1).

N	L	a	E_{initial}	$E_\star$	g_a	B_{geom}
7	6	1.00	43.95516354	9.07×10^{-16}	1.99×10^{-9}	0
9	6	0.75	54.15076270	5.33×10^{-17}	3.91×10^{-9}	0
11	6	0.60	61.09804327	8.07×10^{-16}	3.17×10^{-9}	0
9	8	1.00	43.16126382	1.77×10^{-14}	1.84×10^{-8}	0
11	10	1.00	42.97909652	2.04×10^{-15}	2.13×10^{-9}	0

6.2 Admissible-component guard

At $N = 7, L = 6$, every accepted guard endpoint has three-target $B = 1$, yet each search terminates on its floor with nonzero gradient (Table 2). The monotone fall of E and g_a as the floor is lowered is evidence that the descent is driven toward the dislocation boundary, not toward a resolved interior critical point. This is a controlled numerical result, not a proof that no other metastable critical point exists anywhere in the component.

Table 2: Admissible-component descent at $N = 7, L = 6$.

δ_{floor}	steps	$E_\star$	g_a	final δ_a	B_{geom}
0.10	21	28.63959591	0.47292019	0.100000002	1
0.05	25	26.04401742	0.43767430	0.050000002	1
0.02	28	24.72976823	0.41784577	0.020000002	1

6.3 Artificial fixed-core control

The free residual passes the 2×10^{-6} tolerance on every artificial fixed-core grid, while the full unanchored residual does not (Table 3). This separation is why the result is never promoted to full stationarity.

Table 3: Artificial controls and lowest same-grid spectra.

N	L	free g_a	full g_a	B_{FD}	λ_0^{n+s}	λ_0^{Δ}	λ_0^{A}	$\sigma_{\min} D_W$
7	6	1.45×10^{-8}	0.5792	0.16647	2.26434	1.52463	0.80385	1.50522
9	6	3.33×10^{-8}	0.6631	0.34779	1.80433	1.53862	0.81195	1.46312
11	6	6.11×10^{-8}	0.6379	0.47754	1.50021	1.55084	0.81572	1.43240
9	8	9.46×10^{-9}	0.3453	0.16638	1.16107	1.21935	0.45672	1.44662
11	10	1.02×10^{-8}	0.2362	0.16644	0.75628	1.07593	0.29366	1.42132

The largest exact restricted $n + s$ matrix occurs at $N = 11$: it has 2,835 degrees of freedom and 77,031 nonzeros. On that grid the real diquark block has 1,458 degrees of freedom and 9,234 nonzeros, the gauge block has 2,187 and 13,851, the complex ghost has 729 and 4,617, and the fermion normal operator has 5,832 complex degrees of freedom and 190,432–190,636 nonzeros over the two $N = 11$ backgrounds.

The $N = 7, L = 6$ artificial solve was independently repeated; canonical JSON serialization gave the identical digest

`27c5e4f06d5a27dc8866bd8bad32c8205a0d5f9d7ada10a9eb1dfecb6c4ca6c8`.

The complete machine-readable values, source hashes, and guard traces are in `route_f/output/ap_e7_discrete_rerelax_superhessian.json`.

Fail-closed gate 6.1 (What the fixed-core spectrum means)**.** Positive eigenvalues of $H_{n+s}^{\mathrm{core\ restricted}}$ establish only local stability under variations that vanish on the artificial core and outer boundary. They do not establish a full-grid $B = 1$ stationary point. The smooth derivative estimator B_{FD}

is also not integer on these coarse grids; it is a convergence diagnostic, never a replacement for admissible geometric degree.

7 Fail-closed conclusion and next constructions

The finite-site theorem and numerical unwind close an ambiguity left by AP-E6: simply re-running a generic optimizer cannot produce a genuine topological lattice soliton for the declared naive action. At least one new ingredient is necessary. Three controlled options are:

1. add an infinite or systematically removable barrier $V_{\mathrm{adm}}(\delta_T)$ and prove observables become barrier-independent;
2. replace the naive Skyrme cell by a topology-preserving spherical-volume action and repeat the a, L scan;
3. retain the continuum-relaxed soliton, but discretize only its fluctuation operator with a convergence theorem that never promotes finite-site degree to an exact sector.

Only after one construction produces an unanchored stationary sequence may the fermion contribution be upgraded from $D_W^\dagger D_W$ to the bosonic second derivative of the regulated determinant and combined with interacting SU(2) gauge–ghost cross blocks.

Accordingly the genuine relaxed-$B = 1$, full unanchored stationarity, physical aggregate super-Hessian, four-dimensional dynamics, HMC, quantum continuum, physics-promotion, and portal-start flags are all false.

References

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